

Authentication & Login Blueprint

Livestock Management System

1. Overview

This document outlines the authentication, access control, and user session management for the Livestock Tracking Application. Designed to handle varying levels of rural connectivity and strict data isolation across different farms and user roles.

2. Role-Based Access Control (RBAC)

The system enforces strict multi-tenancy. A user must belong to a 'Farm' or 'Organization' to access records. Below are the primary access roles:

Role	Access Level	Restrictions
Farm Owner	Full Administrative Access	Restricted strictly to their own farm(s).
Farm Manager	Operational Management	Cannot modify billing or overarching system configs.
Worker	Field Data Entry & Task Mgt	Can only view assigned tasks, scan animals, and input data. No financial access.
Veterinarian	Medical & Health Records	Read/Write access to health history and treatments. No financial access.
Auditor / Buyer	Read-Only Traceability	Can only view specific animal records assigned for sale or audit.

3. Authentication Mechanisms

Standard Login (Managers / Owners / Vets):

Requires Email/Phone and Password. Uses secure hashing and JWT (JSON Web Tokens) for session management.

Worker / Field Staff Login:

Simplified access using Phone Number + SMS OTP (One Time Password) or a shared Farm Kiosk PIN for rapid field entry. Designed to minimize login friction in muddy or rushed environments.

Offline-First Resilience:

Once a user logs into the mobile app, a long-lived secure token is stored locally. If the farm loses internet connectivity, the worker remains authenticated locally and can continue recording treatments, weights, and offline scans. Data synchronizes automatically when the connection is restored.

4. Security & Audit Protocols

- **Audit Trails:** Every login event, permission change, and sensitive data export is logged immutably with a Timestamp, User ID, and IP/Device Metadata.
- **Device Revocation:** Farm owners can instantly revoke sessions for lost devices to protect animal ownership and financial records.

- **Data Isolation:** JWT tokens securely encode the *farm_id* to ensure database queries naturally filter out data belonging to other farms.

Prepared for Development Team - Sensitive API keys and database endpoints excluded.